

AN EXAMINATION OF CIGARETTE POLLUTION AND TOXICITY

An Examination of Cigarette Pollution and Toxicity

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28 November, 2024

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The world has been facing serious issues with improper waste disposal and its environmental impacts in the last few decades, and it has become a major concern lately. In fact, 90% of US residents consider pollution a major or minor problem in their state. Cigarettes are contributing to this issue on a daily basis all around the globe as they are and have been the number one littered item for many decades. In a 2020 litter study conducted by Keep America Beautiful, they found that cigarette butts made up 88% of litter that is 4 inches or smaller in the US. In a 2009 study, researchers found that almost 37% of trash collected was cigarette butts (approx. 19 billion butts). Participants then found nearly 10 billion from both roadways and waterways in 2020. Although the average number of cigarettes littered in the US has declined drastically since 2009, this still poses a serious threat to our environment. The Ocean Conservancy's most recent annual beach cleanup event was estimated to have around 486,045 volunteers who collected just over 14 million items. Of that waste, the largest group (25%) consisted solely of cigarette buds [1,2]. Consider what we could accomplish if just 5% of the global population contributed to waste removal efforts! Globally, it is estimated that around 6 trillion cigarettes were produced annually in 2015. In 2025, the number has supposedly reached 9 trillion, and just about 766,571 metric tons of that gets improperly discarded per year [3].

The agricultural side of growing tobacco also contributes in a vast and growing fashion to climate change, deforestation, and chemical pollution in the surrounding environments. In addition to the littering aspect of this situation, cigarettes are an easily avoidable cancer-causing item that kills nearly 7 million people per year [4]. In the early 1950s, scientists began to discover cancer and disease-causing agents in the smoke produced by cigarettes. This prompted the tobacco industry to sell their cigarettes with filters to reduce the measured amount of tar in the smoke, which 99% of cigarettes sold in the US now include. These filters are composed of

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nearly 15,000 fibers of this nonbiodegradable substance called cellulose acetate, which in its nature is classified as plastic [5]. The main issue (for the environment) with these filters is that they are strategically used to absorb and trap different harmful constituents, which are usually the most harmful ones found throughout the cigarette, that are transformed from the tobacco leaf to cigarette smoke so they do not continue into the user's lungs [6]. Not only are cigarette butts globally a significant source of microplastics/nanoplastics (MPs/NPs), but they contain thousands of carcinogenic chemicals and toxicants, including heavy metals, polycyclic aromatic hydrocarbons (PAHs) as well as amines, and nicotine.

Despite being a social norm for decades, the issue of cigarette butt waste is becoming increasingly apparent as we grapple with global environmental challenges. With over 6 trillion cigarettes being produced annually [3], cigarette butts are a significant source of MPs and NPs that can persist in and threaten many different environments for an unknown period of time. The low density of a cigarette butt allows it to travel easily throughout cities, eventually reaching its preferred destinations such as storm drains, waterways, and beaches. Even under harsh environmental conditions, these filters can persist for a long time, slowly releasing more synthetic fibers [7]. This long-term persistence underscores the need for sustainable solutions to this problem.

In order to be decomposed by organisms, cellulose acetate needs to be de-acetylated by high UV radiation and chemical hydrolysis [8]. However, perfect conditions like that are unlikely and are mainly achieved through controlled research environments. For example, a 20-40% loss of mass was observed after 2 years of being exposed to the sun on a flat roof in Tennessee [9] rather than a cigarette butt exposed to a tropical beach environment (not a control test) for 5 years losing about 30-35% of its original mass. After the early decay of the small labile fraction,

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corresponding to the external paper layer and residual tobacco, degradation proceeds very slowly, regardless of the environmental conditions [7,8]. With respect to sandy environments, cigarette butts that have accumulated over time will become less likely to be found by cleanups and will undergo disintegration due to physical conditions rather than degradation due to chemical conditions. This will generate an enormous amount of synthetic fibers and release heavy metals that will persist in the environment for an unknown period of time [7]. MPs and NPs can enter the human body via inhalation, absorption, or ingestion and, from there, as documentation states, can bioaccumulate in multiple different organs including reproductive organs and the brain [10]. A study was conducted on Zebrafish embryos to examine the effects of exposure to MPs and NPs, as these outcomes have not been well elucidated for many organisms. Exposure to NPs alone caused early-stage death, while exposure to both MPs and NPs resulted in pathological angiogenesis, developmental disorders, and growth impairment [11].

Due to anthropogenic actions, nanoscale contaminants and particulate matter have gained abundant attention in the last few decades as nanotechnology has advanced. As nanoscale particles (NSPs) travel through air and water, they can pick up not only pick up many toxins, chemicals, and pollutants along the way, but also metals and organic contaminants. The biggest issue about NSPs is their ability to pass through various protective layers like skin or cell membranes and carry the particles they have picked up. Different metals like As, Pb, Cd, Cu, Ni, Cr, Co, Al, Mn, Zn, and Fe attach themselves to the NSPs of carbon released from combustion, thus getting caught in the filter. After soaking CBs in rainwater, Fe, Al, and Zn were the most predominant metals that leached from the CB. Metals that were also released but less prevalent include Cr, Mn, Cu, Pb, Cd, As, and Co. Given the amount of CBs littered yearly, it equals a tremendous amount of dissolved and nanoscale metal fractions released into the environment

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annually [12]. Another study was done regarding CBs and their leachates in rainwater and found that the release of the metals Ba, Fe, Mn, and Sr could continue for up to or longer than a month. On the contrary, metals like Zn, Pb, Ti, and Ni release all of their leachable particles within a day, which could cause acute biological effects to local organisms that are sensitive to those metals [13].

In most studies, it is imperative to notice that CBs with remnant tobacco contain much higher levels of toxins than a smoked CB with no tobacco, as well as an unsmoked CB. With remnant tobacco comes the inclusion of the incomplete burning of the leaf that can contain large amounts of PAHs. The awareness of PAHs has increased dramatically due to their potential bioaccumulation and carcinogenic/mutagenic properties. A study done by Dobaradaran, S. et al. concluded that, annually, close to 5 tons of PAHs could potentially be released into water systems, which completely exceeds the Water Framework Directive standards and safe drinking water recommendations. Naphthalene (1.3 tons) and fluorene (1.6 tons) are both lightweight PAHs and were the most prevalent PAHs found in the study. Although, this study was conducted in a 21 day time frame and CBs spending more time in the environment may release more heavy weight PAHs. An influx of PAHs will induce a high potential for adverse effects on aquatic organisms as well as human health [14]. Aromatic amines (AA) also pose another risk to aquatic organisms, mainly speaking sensitive crustaceans and fish. These AAs can lead to mutations during DNA replication and disrupt genetic stability if tracked into the body of an organism. The AA named aniline had the highest leaching rates in multiple tests with distilled water and river water (both distilled and river water had the same results) [15]. Aniline can induce acute and chronic effects on organisms as it replaces the hemoglobin in blood with methemoglobin, which does not transport oxygen effectively. A 2003 study showed that fish can absorb and metabolize

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aniline and, through tests, was found to have adverse effects on tilapia even at concentrations as low as 0.02mg/L. Observing conditions at 90 days with these low concentrations, the fish's growth and food conversion were less efficient, and a reduction appeared in reproduction yield. Generally, aniline can impede the reproductive functions of fish at a concentration of 0.05mg/L. A higher concentration of around 2.65mg/L can also drastically affect the phytoplankton population in water. What is even more disturbing about this is that the NPs produced by CBs are colloidal particles that AAs can adsorb to and hence become suspended in the aquatic environment for a more extended time than without the NPs. Both crustaceans and fish were shown to be affected by this AA [16].

The leachates from smoked CBs were also reported to have acute toxic effects on saltwater fish like topsmelt and freshwater fish like fathead minnows. For both species, it was found that an LC50 of around 1 CB/L was obtained, with the LC50 getting slightly larger for CBs with no tobacco remnants. Marine bacteria had a similar sensitivity to smoked CBs with remnant tobacco as the topsmelt and minnows. This study also stated that previous studies obtained EC50 concentrations much lower than those for the topsmelt and minnows for the daphnid, *Ceriodaphnia cf dubia*. Smoked CBs with remnant tobacco were acutely toxic to these daphnids at a concentration of around 0.03-0.08 CB/L. The toxins leaching from the CBs that affected the daphnids and marine bacteria the most were pesticides since pesticide residue on tobacco is not regulated, and nicotine, a very toxic form of insecticide [17]. Nicotine as a leachate poses a serious threat to not only aquatic life but human life as well. A study executed in Yasuj, Iran, by Mohammadi et al. showed the importance of knowing how much nicotine leaks into runoff and storm drains from CBs. With a population of around 119,000, the best and worst case scenarios presented themselves as 335,070 and 481,950 gr/year of nicotine being leached

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into their water resources. When rain hits the city, it is estimated that the yields should be about 27.5% more than what was previously noted. Many citizens in the city obtain their water through groundwater sources, so it is an issue that needs an immediate solution [19]. This data was retrieved from just one city; therefore, it is hard to imagine the amount of adverse effects this could lead to worldwide.

A study done by Shah et al. indicated that CBs also pose a serious threat to land organisms. This study specifically concentrated on the effects CBs had on *Azolla pinnata* as this fern species has been increasingly used for various purposes, mainly because it is a hyperaccumulator. Being in the presence of CB leachates for only 6 days, *Azolla p.* had shown adverse effects to its surface area, root length, and amount of fronds. Its whole photosynthetic process was drastically impaired as the plant became less efficient in converting light into usable energy. The lethal dose value obtained from the study demonstrates that the plants could not sustain themselves with a concentration of around 20-30 CBs [18]. Various leachate concentration tests were also used to assess their impact on intertidal mollusks and possibly other intertidal organisms. The full leachate concentration was 5 CB/L with a soak time of 2 hours over 8 days. The mortality rate for all species tested was, unfortunately, 100%. That said, further testing revealed that mortality rates were species-specific. When one species gets affected more than others, it could disrupt the dynamics of their environment and potentially cause issues due to less biodiversity [20].

Similarly, a study reported that CB leachates heavily affected another daphnid, the cladoceran. *Ceriodaphnia cf. dubia* and *Vibrio fischeri* were tested in this project, and observations showed considerable reaction variation. The *V. fischeri*, a bioluminescent bacteria, was evaluated using a microtox test. The EC50 values for *C. cf. dubia* varied between 8.9-25mg

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CB/L between the least and most toxic CBs, while *V. fischeri* had an exponentially larger gap at 104-832mg CB/L for its EC50. The researchers noted that nicotine and ethylphenol were the primary toxicants that affected *C. cf. dubia*. This shows that CBs contain such a complex and diverse amount of toxins; the effects its leachates can have on one species should not and cannot be used to speculate the effects it may have on another species [21]. Conducting an array of further studies to determine the effects of CBs on individual species is very pertinent to figuring out possible solutions in the future.

The research presented in the last couple of decades should be a convincing call to action for the world population about how cautious we must be about the short-term and long-lasting effects CBs might impose on aquatic and terrestrial ecosystems and human health. More studies need to be conducted in order to obtain a broader lens through which to convince others who might not have been swayed by what the scientific studies have already presented. Recognition that cellulose acetate cigarette filters have been nothing more than a marketing tactic with no actual benefit to the health of smokers is as important as finding the best ways to mitigate CB litter in the urban environment as they are an environmental blight. With the knowledge of how cigarettes can have a negative effect on human health being a globally known fact, it seems possible to curb the uptake of smoking by youth as well as promote successful quitting by altogether banning the production and sale of cigarettes with filters [22]. As for the littering portion of the issue, labeling cigarettes as single-use plastics and requiring tobacco companies to throw a prominent warning on the cigarette packages not to litter the package or product could decrease the average littering rates nationwide. As far as policy change, it would also be of great help to enact a “bottle bill” on the CBs, as states that have enacted this for glass bottles or metal cans have seen a considerable reduction in those littered items [5]. Keep America Beautiful

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found, in an aggregate count, 219 million littered cans/bottles per capita with instated bottle bills as opposed to 1.2 billion cans/bottles per capita with no bottle bills [1]. It can act as a fantastic way to provide money for those who are in need of it, as well as help to take these dangerous CBs off of the ground and into a proper recycling system.

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